

UCS714: Agentic AI				
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Course Objective: The course aims to develop foundational understanding of Generative AI and Large Language Models, introduce the principles and architectures of Agentic AI systems, train students in prompt engineering for autonomous agents, enable the design and development of single and multi-agent systems, and equip learners with evaluation, optimization, and governance techniques for responsible deployment of agentic AI solutions.				
Foundations of Generative and Agentic AI: Overview of Generative AI; fundamentals of Large Language Models and their capabilities and limitations; applications of Generative AI across domains; comparison of traditional AI and Agentic AI; components of intelligent agents including goals, memory, reasoning, and action; autonomy and feedback mechanisms in agent systems.				
Prompt Engineering and Reasoning Strategies for Agents: Role prompting; zero-shot and few-shot prompting; Chain-of-Thought prompting; ReAct prompting; self-reflection prompting; system prompts versus user prompts in agent architectures; prompt structuring for multi-agent interactions; prompt injection risks and prompt hygiene practices.				
Agent Architectures and Design Paradigms: Characteristics of intelligent agents including autonomy, reactivity, proactiveness, and social ability; reactive and deliberative agent models; reflection design pattern; tool-use pattern; planning agents and task decomposition; multi-agent collaboration architectures; architectural considerations for robust and scalable agentic systems.				
Collaborative Multi-Agent Systems and Workflow Orchestration: Agent communication models including direct and mediated communication; structured and unstructured message passing; shared memory models such as blackboard systems; DAG-based workflow orchestration; recursive delegation and feedback-driven iteration; interoperability protocols including Model Context Protocol (MCP) and Agent-to-Agent (A2A) protocol; secure cross-agent communication mechanisms.				
Deployment and Evaluation of Agentic AI: Workflow orchestration using low-code tools such as N8N; triggers, webhooks, and automation nodes; API integration, authentication, and secure token management; integration with databases and external systems; deployment of scalable multi-agent pipelines; error handling, fallback mechanisms, and retry strategies; performance metrics including task success rate, latency, reasoning depth, response coherence, and cost tracking; prompt optimization and versioning; debugging and monitoring of agent behavior.				

Lab Experiments - Exploring LLM APIs and prompt testing, implementing goal-driven single agents, building reasoning-based agents, designing tool-using agents, implementing planning and reflection agents, multi-agent collaboration experiment, Building agent workflows with automation tools, API integration and deployment, Performance benchmarking and evaluation

Course Outcomes (COs)

At the end of the course, the student will be able to:

CO1: Explain the principles of Generative AI and Agentic AI systems.

CO2: Apply prompt engineering techniques for reasoning and autonomous behavior.

CO3: Design and implement agentic AI architectures using standard design patterns.

CO4: Develop end-to-end multi-agent pipelines integrating tools and APIs.

CO5: Evaluate agentic AI systems using performance metrics

Text Books

1. "Multi-Agent Systems: Algorithmic, Game-Theoretic, and Logical Foundations" – Yoav Shoham and Kevin Leyton-Brown.
2. Mastering AI Agents: From Design to Deployment of Autonomous Systems by Elbert Gale, ISBN-13: 979-8307403822
3. Building AI Agent: Learn Building AI Agents, A Practical Guide for Building Intelligent Agents by Natenapis Faraksa.

Reference Books

1. "Designing Autonomous Agents: Theory and Practice from Biology to Engineering and Back" – Pattie Maes.
2. Build Your Own AI Agent: A Hands-On Guide for Beginners by Sujai G, ISBN-13: 979-8306885971